

Gender Bias in AI Hiring Systems and Its Impact on Women in STEM Careers

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CST462s: Race, Gender, Class in the Digital World

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Spring 2026

Team 7

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Research Problem

Artificial intelligence (AI) is being used more often in hiring and recruiting. Many companies now use automated systems to scan, rank applicants and decide which applicants to move forward in the hiring process. These tools are meant to make hiring faster and help companies deal with large numbers of applications. However, some critics have raised concerns that these systems may repeat or reinforce already existing inequalities in the workforce.

This issue is especially important in STEM fields. Women have historically been underrepresented in science, technology, engineering and math careers. Even though there have been efforts to increase diversity, women still face barriers when entering these fields. AI hiring systems are often trained using past hiring data, and if those past decisions favored male candidates, the algorithm will then learn those same patterns. This could cause the system to rank male applicants higher than female applicants even when they have similar qualifications.

The social impact of this issue could be significant. If AI hiring tools contain bias they could limit opportunities for women trying to enter STEM careers and continue the gender gap that already exists in these industries. This project will examine how AI driven hiring algorithms may affect employment opportunities for women in STEM and explore whether these technologies contribute to gender inequality in hiring.

Literature Review

Fabris et al. (2023) examine how Artificial Intelligence (AI) is being used more in hiring to help organizations to handle large numbers of job applications and to make the

recruitment process more efficient. Many companies now use algorithmic tools to screen resumes and to rank applicants. While these systems are often described as objective and efficient, researchers warn that they can reproduce existing social inequalities. Fabris et al. (2023) examine how algorithmic hiring systems can introduce bias when they rely on historical employment data.

According to Fabris et al. (2023), many AI hiring systems are trained using data from past hiring decisions. Because this data may reflect earlier patterns of discrimination or unfair representation the algorithms will learn to favor characteristics of candidates who were previously hired. In fields such as STEM where men have historically been overrepresented, this can disadvantage women applying for similar roles. If the algorithm learns patterns based on past hiring trends it could unintentionally rank male applicants higher than female applicants with similar qualifications.

Bias can also be found in the factors the algorithms evaluate during the hiring process. Automated systems typically prioritize certain keywords, education backgrounds or career experiences. The prioritized keywords may be more common among the already dominant group in the field. Some AI hiring tools also analyze video interviews, which can introduce additional bias (Fabris et al., 2023).

Another issue discussed by Fabris et al. (2023) is the lack of transparency in many algorithmic hiring systems. Because these tools are often developed by third-party companies it can be difficult to understand how decisions are made. This makes it difficult to identify or correct biased outcomes. The authors suggest that fairness testing and oversight are important to help ensure AI hiring technologies do not reinforce the already existing inequalities.

Similarly, Chen (2023) argues that AI is an excellent tool and resource that can handle various tasks and automate processes. The use of AI boosts efficiency, and due to its capabilities, many companies in the hiring and recruiting process are able to save time and review applicants more quickly. Nevertheless, the use of AI in recruitment raises concerns about ethics and rightfulness because models in the process of training may have used past data, which often underrepresents women and favors male applicants. According to Chen (2023), AI is not itself inherently biased, but bias stems from the people who originally trained and designed these systems. The IT and technology industries are predominantly male-dominated. Therefore, AI systems developed on this data may unintentionally prefer male applicants over women.

As discussed earlier, AI is trained using past data, which can often be discriminatory toward certain gender and racial groups. As reported by Chen (2023), datasets used in AI tend to lean toward historically conventional groups, which may result in underrepresentation of women in the decision-making process. This raises the question of the equality and reliability of AI in the recruitment process. Considering all these factors, AI systems may make decisions based on existing biases, which will further amplify issues related to gender and race inequality in the hiring process.

To further examine cases where AI may not make justified decisions, the article discusses an incident of a hiring tool failing to do its job in an unbiased manner. Chen (2023) verifies this by stating how in 2014 Amazon used an ML machine in the hiring process with the hope that it would be gender-neutral; however, it failed to do so. This strongly suggests that AI is more likely to use algorithms that may discriminate against

women candidates automatically due to the already established lack of women in STEM careers.

As AI continues to advance and more tasks get automated, it is clear that it will be actively used in the hiring process. At the same time, due to the past biased nature of datasets which AI uses, its determinations may need careful reconsideration. Chen (2023) suggests that people should be aware of how AI makes its decisions and the motives behind them as a way to prevent favoritism toward male applicants over women. More transparency within the AI hiring process allows companies to see where exactly it lacks to make bias-free decisions and correct them.

AI is becoming more common in hiring because it can quickly review large numbers of job applications and help companies save time. However, researchers have raised concerns that these systems may not be as fair as they appear. Albaroudi et al. (2024) explain that AI hiring tools are designed to identify patterns in data, which means they often rely on information from past hiring decisions. While this can improve efficiency, it also creates a risk that existing inequalities in the workforce will be repeated through automated systems.

One major issue is that historical hiring data may reflect long-standing gender imbalances, especially in fields like science, technology, engineering, and mathematics (STEM). Because men have traditionally been overrepresented in these areas, the data used to train AI systems may favor characteristics associated with male candidates. As a result, qualified women may be ranked lower or screened out before human reviewers ever see their applications. The authors note that this problem is not caused by the technology itself but by the biased information used to train it.

The study also points out that algorithms often evaluate factors such as education history, career progression, and specific keywords on resumes. These criteria can unintentionally disadvantage women who may have taken time away from work for caregiving or who followed nontraditional career paths. In addition, many hiring algorithms operate as “black boxes,” meaning employers and applicants cannot easily see how decisions are made. This lack of transparency makes it difficult to identify or correct unfair outcomes.

Overall, Albaroudi et al. (2024) conclude that AI has the potential to improve hiring processes, but without careful oversight it can reinforce existing social inequalities rather than reduce them. The authors emphasize the importance of using diverse training data, regularly testing systems for fairness, and increasing transparency in automated decision-making. Their findings suggest that organizations must be cautious when using AI in recruitment to ensure it does not limit opportunities for women pursuing careers in STEM.

While a lot of research talks about how AI hiring systems can create bias during screening and decision-making, there is also another important issue that is not talked about as much. This is how these systems affect people before they even apply. In particular, the awareness or perception of bias in AI hiring systems can influence whether women decide to apply for jobs in the first place, especially in competitive, male-dominant fields like STEM.

The study by Ip (2025) looks at this idea using an experimental setup where participants had to decide whether or not to apply for a job under different AI hiring conditions. Some of these conditions included systems that were biased against women

and others that tried to reduce bias using different fairness methods. The results showed that when women were told that the AI system was biased against them, many qualified female candidates were less likely to apply. Specifically, women became 18.1 percentage points less likely to apply for the competitive position when informed of bias favoring men, while men became 10.9 percentage points more likely to apply (Ip, 2025). This means that the impact of AI bias is not only happening during hiring decisions, but even earlier, when people decide whether to enter the process at all.

Another important finding from the study is that men reacted differently. When the system favored men, they were actually more likely to apply. This creates an imbalance where men continue to enter the applicant pool while women may choose to opt out. Over time, this can make the problem worse because fewer women are applying, which can then lead to fewer women being hired.

STEM fields are still mostly male-dominated, and women already deal with a lot of challenges when it comes to entering these fields. For example, there are fewer role models, and many women may feel less confident in their abilities or experience pressure from negative stereotypes about their performance. When you add the idea that an AI hiring system might be biased, it can make the situation even worse. Women may start to feel discouraged before even applying. This leads to fewer qualified women entering the applicant pool, which keeps the gap in STEM from improving. Because of this, companies cannot just focus on fixing the algorithm itself. They also need to be clear and transparent about how their systems work and show that they are fair, so applicants feel more confident applying.

If AI hiring systems are seen as biased or unfair, this may discourage even more women from applying. So this will result in fewer women applying, fewer women being hired, and future AI systems continue to learn from this unequal data.

The study also shows that even when companies try to fix bias in AI systems, it may not fully solve the problem. If people do not trust the system, they may still choose not to apply. This means that improving fairness in AI is not just about changing the algorithm, but also about making sure people understand and trust the system.

Overall, this study shows that AI hiring systems can impact women in STEM not only through biased decisions but also by discouraging them from applying in the first place. This makes the problem more serious because it affects the entire hiring process, not just the final decision.

Across the reviewed sources, a consistent theme is that AI-driven hiring systems have the potential to reinforce existing gender inequalities rather than eliminate them. All three studies emphasize that these technologies are often trained on historical hiring data, which reflects past workforce imbalances. Because men have traditionally been overrepresented in STEM fields, algorithms may learn to associate male candidates with success and unintentionally favor them during screening and ranking processes. This creates a risk that qualified women will be overlooked, even when they possess similar skills and experience.

Another common finding is that bias can arise from the criteria used to evaluate applicants. Automated systems frequently analyze factors such as education background, work history, and resume keywords. While these metrics appear objective, they may advantage candidates who follow traditional career paths that are more

common among men in STEM fields. Women who have taken career breaks, changed fields, or pursued nontraditional educational routes may therefore be penalized. Some sources also note that newer tools, such as video interview analysis, introduce additional opportunities for bias through facial recognition, speech patterns, or behavioral assessments.

A further shared concern among the sources is the lack of transparency in many algorithmic hiring tools. Because these systems are often proprietary, employers and applicants may not understand how decisions are made or why certain candidates are rejected. This makes it difficult to identify discriminatory outcomes or hold organizations accountable. Without clear oversight and fairness testing, biased algorithms can influence hiring decisions on a large scale, potentially widening existing gender gaps in STEM employment.

In addition to these concerns, research also shows that AI hiring systems can affect people even before the hiring process begins. Most studies focus on how these systems evaluate candidates, but fewer look at how they influence who decides to apply in the first place. The study by Ip (2025) found that when people were told that an AI system favored men, qualified women were much less likely to apply, while men were more likely to apply in the same situation. This shows that just the idea or perception of bias can discourage women from entering the hiring process at all.

Despite these concerns, the literature also suggests that AI hiring systems are not inherently discriminatory and can be improved through responsible design and regulation. Researchers highlight the importance of diverse training datasets, ethical standards, and ongoing monitoring to reduce bias. Overall, the sources agree that while

AI has the potential to make recruitment more efficient, it must be carefully managed to ensure it promotes fairness rather than perpetuating inequality. These findings support the need for further research into how AI-driven hiring practices affect women's opportunities in STEM fields.

Research Question

How do AI-driven hiring algorithms influence hiring outcomes and application behavior for women in STEM fields?

Research Design

Target Research Participant Audience

- Individuals connected to STEM education or careers who have experience with online or AI-based hiring systems.
- Age: 18 and above.
- Any gender, with emphasis on women in STEM.
- Individuals who have recently applied for jobs using automated or technology-based hiring tools.
- Diverse backgrounds in terms of race, education level, and experience.

Recruitment methods

- Recruiting participants through personal networks and classmates involved in STEM fields.
- Reaching out to university students or alumni in STEM programs.
- Using social media platforms (LinkedIn, group chats) to find individuals with relevant experience.

- Contacting individuals who have recently applied to jobs using online hiring systems.
- Utilizing existing connections within academic or professional STEM communities.

Interview Questions

1. How do you think the use of AI or automated systems in hiring affects fairness for women applying to STEM jobs?
2. Have you ever experienced or observed bias during an online or technology-based hiring process(such as resume screen tools, automated assessments, or video interviews)? If so, how did it affect the outcome?
3. Have you ever felt discouraged from applying to a STEM job because the hiring process seemed automated or based on AI tools? Why or why not?
4. What concerns, if any, do you have about how AI systems make hiring decisions?
5. What factors do you think AI systems rely on when evaluating candidates, and how might those factors impact women in STEM?
6. How do AI or automated hiring systems influence how you prepare for or approach applying to STEM jobs?
7. Do you think AI hiring systems reflect past inequalities in STEM fields, such as underrepresentation of women? Please explain your reasoning.
8. How important do you think it is for companies to be transparent about how their AI hiring systems make decisions?

Interview format

- Interviews will be conducted using a Google Form, where participants can respond to questions asynchronously.
- Each participant will be provided the same set of semi-structured interview questions.
- Participants will complete the interview at their own pace within the given timeframe.
- This format allows flexibility and accessibility for participants who may have varying schedules.
- Responses will be collected and reviewed from the Google Form for analysis.
- All interviews will be completed by the end of Week 5 to allow sufficient time for data analysis.

Participant 1 is a personal acquaintance in their early twenties who is currently pursuing a Bachelor of Science degree and working in a hospital setting. They are involved in a structured clinical program that includes rotations across different departments and exposure to both patient care and administrative responsibilities. They also have experience participating in training and onboarding processes and occasional involvement in research-related activities. This participant provides a valuable perspective from someone actively engaged in the medical field, offering insight into real-world experiences within a STEM environment.

Participant 2 is a personal acquaintance in her early thirties who holds a master's degree in a STEM field. She was recently affected by a mass layoff and has since been

actively seeking to re-enter the workforce in her field. This individual represents an ideal participant for this study because she directly aligns with the target population of women in STEM navigating modern hiring processes. Her advanced education and prior professional experience provide insight into expectations at a highly skilled level, while her recent job loss places her in an active job search context. This allows her to offer current, firsthand perspectives on hiring practices, including interactions with AI-driven or automated systems. Additionally, her career stage as a mid-career professional makes her experiences relevant to broader workforce trends impacting women in STEM.

Participant 3 is an undergraduate student in her early twenties pursuing a STEM-related degree. She responded to the survey through a shared link rather than as a personal acquaintance, offering a more external perspective. She has experience applying to internships and entry-level roles using automated application systems. This participant provides insight into how early-career women in STEM interact with and perceive AI-driven hiring processes.

Participant 4 is a personal acquaintance in her late twenties who holds a bachelor's degree in a STEM-related field and is currently employed as a software developer at Google. She works in a fast-paced, highly technical environment, contributing to the development and maintenance of large-scale software systems while collaborating with cross-functional teams. Having successfully navigated a competitive hiring process, she offers valuable insight into technical interviews, recruitment expectations, and career progression within the tech industry. Her professional experience also provides perspective on the challenges and opportunities faced by

women in software engineering, making her a strong representative of individuals actively engaged in the modern STEM workforce.

Findings

The interview responses show mixed experiences with AI hiring systems among participants in STEM related fields. Two participants said they had not clearly seen gender bias in automated hiring, while one suggested that bias could still exist depending on how systems are designed, and another noted that fairness depends entirely on how the system is trained and monitored (Participant 1, personal communication, April 2026; Participant 2, personal communication, April 2026; Participant 3, personal communication, April 2026; Participant 4, personal communication, April 2026).

Even when bias was not directly observed, all participants pointed out that AI systems tend to rely heavily on keywords, structured resumes, and specific experience requirements, which raised concerns that qualified candidates could be filtered out if their background did not match exactly what the system was looking for. Participants had different reactions to this — one described the process as neutral and efficient, another said they sometimes stop applying when it feels too automated, and another noted that heavy automation can make the process feel impersonal, as though their application is reduced to data points rather than their full story (Participant 1, personal communication, April 2026; Participant 2, personal communication, April 2026; Participant 3, personal communication, April 2026; Participant 4, personal communication, April 2026).

Another recurring theme shared among participants was that most did not feel discouraged from applying to STEM jobs even if the hiring process was automated by AI. Some participants mentioned that automation could lead to faster responses, and they believed that their applications would still eventually reach a human if they passed the initial screening stages (Participant 1, personal communication, April 2026; Participant 2, personal communication, April 2026; Participant 3, personal communication, April 2026; Participant 4, personal communication, April 2026).

Finally, transparency came up in every interview. Participants said it is important to understand how AI systems make decisions, even though there were some concerns about people potentially taking advantage of that information. One participant emphasized that transparency builds trust and accountability and allows companies to be held responsible for identifying and correcting bias.

Overall, AI hiring systems shape both how applicants are evaluated and how they approach applying. The reliance on keywords and structured criteria suggests that these systems may favor certain types of experience over others, particularly disadvantaging women who may have nonlinear career paths or gaps due to caregiving responsibilities. This shows that AI hiring systems are not just part of the decision making stage but influence multiple parts of the hiring process, including how applicants present themselves and whether they continue in the process.

Conclusions

The research examined how AI-driven hiring systems influence hiring outcomes and application behavior for women in STEM. The findings suggest AI has an impact,

though not always in direct or obvious ways. Participants did not report clear cases of gender discrimination, but the results indicate bias may operate indirectly through how systems evaluate candidates and structure the hiring process.

A key takeaway is that AI systems rely heavily on structured criteria such as keywords, job titles, and specific experience. This can filter out qualified applicants who do not match exact patterns, which may disproportionately affect women with nontraditional career paths or employment gaps. This aligns with the literature suggesting AI can reinforce existing inequalities without explicitly targeting a group.

AI systems also shape how applicants approach the job search. Some participants adjusted resumes to match perceived system preferences, while others found the process impersonal or discouraging. Although most were not fully deterred, automation clearly influenced their experience and behavior.

Transparency emerged as an important but complex issue. Participants agreed it is necessary for trust, but had mixed views on how much should be provided due to concerns about system manipulation.

Overall, AI hiring systems influence both outcomes and behavior, but primarily through structural factors rather than direct discrimination.

Recommendations to Solve the Problem

As discussed earlier, AI is not inherently biased; however, it may produce biased judgments due to how it was originally designed by developers. One solution to solve the problem of AI's potential biased decision-making in the hiring process of women is for companies to be willing to closely monitor the outcomes of AI decisions. To further

explain, if women are being rejected or dismissed more during the hiring process compared to male applicants, the company should investigate closely to determine whether the AI is making biased decisions due to the past data it is using, which often underrepresented women and favored male applicants. Subsequently, if bias is present in the system, the company should strongly reconsider using that system in the hiring process to prevent gender-based discrimination in the workplace. In addition, companies should implement regular audits of their AI hiring tools, such as quarterly evaluations that compare hiring outcomes across gender groups. These audits can help identify patterns that may not be immediately visible and allow organizations to correct issues early. If bias is detected, companies can retrain their models using updated and more balanced data or adjust the criteria the system uses to evaluate candidates. By actively monitoring and improving these systems, organizations can reduce the risk of reinforcing existing inequalities in STEM hiring.

Another important recommendation is to improve the quality and diversity of the data used to train AI hiring systems. Since these systems learn from past hiring decisions, they often reflect historical patterns where men have been overrepresented in STEM fields. To address this issue, companies should ensure that their training data includes a wider range of applicants, including women with different educational backgrounds, career paths, and experiences. This includes individuals who may have taken career breaks, switched fields, or followed nontraditional paths. By incorporating more diverse data, AI systems can better recognize different forms of qualification and reduce the likelihood of favoring only traditional career trajectories. Additionally, developers should carefully review which factors the algorithm prioritizes, such as

keywords or specific job titles, to ensure these criteria do not unintentionally disadvantage certain groups. Improving the design and training of AI systems can help make hiring processes more equitable and inclusive.

Increasing transparency in AI hiring systems is another key solution. Many participants in this study emphasized the importance of understanding how AI makes decisions, as transparency helps build trust and accountability. Companies should clearly communicate when AI is being used in the hiring process and provide general information about how candidates are evaluated. For example, they can explain what types of qualifications, skills, or experiences the system considers most important. This does not require revealing the entire algorithm, but it allows applicants to better understand how to present themselves and feel more confident in the fairness of the process. Greater transparency can also encourage more women to apply for STEM positions by reducing uncertainty and concerns about hidden bias. At the same time, companies should balance transparency with safeguards to prevent applicants from manipulating the system, ensuring that fairness is maintained.

At a broader level, policy intervention is necessary to address this issue as a systemic problem. Individual companies may take steps to reduce bias, but without industry-wide standards, these efforts may be inconsistent. Governments or regulatory bodies should establish guidelines that require companies to regularly test AI hiring systems for bias and ensure they meet ethical standards. This could include mandatory reporting of hiring outcomes by gender, as well as independent audits of AI systems used in recruitment. Such policies would hold organizations accountable and encourage the development of fairer technologies across industries. By creating clear standards

and oversight, policy-level solutions can help ensure that AI hiring systems promote equal opportunities rather than reinforce existing inequalities.

Finally, companies should ensure that human oversight remains part of the hiring process. While AI can improve efficiency by screening large numbers of applications, it should not be the sole decision-maker. Human reviewers can provide context that AI systems may miss, such as leadership potential, adaptability, or the value of nontraditional experiences. Incorporating human judgment at later stages of the hiring process can help prevent qualified candidates from being unfairly filtered out early on. This approach combines the efficiency of AI with the critical thinking and fairness of human decision-making, creating a more balanced and inclusive hiring system overall.

Recommendations for Further Research

This study was limited by a small sample size of only four participants, which means the findings cannot be generalized to all women in STEM. Future research should include a larger and more diverse group of participants across different industries, experience levels, and backgrounds. This would help provide a broader understanding of how AI hiring systems affect different groups of people.

Another area for future research could explore how different types of AI hiring tools (such as resume screening vs. video interview analysis) impact applicants differently. Since AI is used in multiple stages of the hiring process, understanding where bias is most likely to occur could help target improvements more effectively.

References

- Albaroudi, E., Mansouri, T., & Alameer, A. (2024). A comprehensive review of AI techniques for addressing algorithmic bias in job hiring. *AI*, 5(1), 383–404.
<https://doi.org/10.3390/ai5010019>
- Chen, Z. (2023, September 13). Ethics and discrimination in artificial intelligence-enabled recruitment practices. *Humanities and Social Sciences Communications*, 10. <https://doi.org/10.1057/s41599-023-02079-x>
- Fabris, A., Baranowska, N., Dennis, M. J., Graus, D., Hacker, P., Saldivar, J., Borgesius, F. Z., & Biega, A. (2023). Fairness and bias in algorithmic hiring: A multidisciplinary survey. *arXiv*. <https://doi.org/10.48550/arXiv.2309.13933>
- Ip, E. (2025). Fair AI in hiring: Experimental evidence on how biased hiring algorithms and different debiasing methods affect the quality and diversity of applicants. *Behavioral Science & Policy*. <https://doi.org/10.1177/23794607251353585>

Appendix A (Angelica)

Participant 1 Interview Responses

Gender	Age Range	Connection to STEM	Applied Using Automated System
Woman	18-24	Undergraduate student	Yes

Question 1: How do you think the use of AI or automated systems in hiring affects fairness for women applying to STEM jobs?

Participant 1: I think it does not offer unique benefits due to the gender. I feel like both males and females are treated the same by AI and they have the same opportunities and go through the same process.

Question 2: Have you ever experienced or observed bias during an online or technology-based hiring process (such as resume screen tools, automated assessments, or video interviews)? If so, how did it affect the outcome?

Participant 1: No, I never noticed bias. I am using handshake and it has some kind of automation and never noticed bias or differentiation of filters.

Question 3: Have you ever felt discouraged from applying to a STEM job because the hiring process seemed automated or based on AI tools? Why or why not?

Participant 1: No, because the endpoint was always people behind it and the final decision was them. The only problem was sometimes they used AI notetakers as a helping aid which sometimes took a little bit wrong notes but I still had people talking with me and forming impression about me.

Question 4: What concerns, if any, do you have about how AI systems make hiring decisions?

Participant 1: Sometimes I noticed standardized decisions about people. I noticed that it makes decisions based on the presence of certain keywords but sometimes they could hire people who does not have required skills but would be better fit because of other skills and experiences that they can bring to the team.

Question 5: What factors do you think AI systems rely on when evaluating candidates, and how might those factors impact women in STEM?

Participant 1: I think they really on certain keywords. I think employers use keywords which gives an algorithm to AI to decide. I noticed that I am offered jobs which is irrelevant to my major but the employer mentioned my major as a keyword to have more candidates to apply and sometimes I see that relevant jobs doesn't hire me because my major is not used as a keyword.

Question 6: How do AI or automated hiring systems influence how you prepare for or approach applying to STEM jobs?

Participant 1: I think it gives an idea how to construct the resume. Never know the process is not using a lot of cognitive abilities AI is making decisions based in the simple algorithms and that makes to understand the logic and makes you stand out.

Question 7: Do you think AI hiring systems reflect past inequalities in STEM fields, such as underrepresentation of women?

Participant 1: I never noticed that when applying to jobs

Question 8: How important do you think it is for companies to be transparent about how their AI hiring systems make decisions?

Participant 1: I think it is really important. Because currently employers are gatekeeping the algorithms and that makes them to lose really talented candidates. People who

have great skills stays in the shadow when some people who were just “lucky” to understand the logic accidentally get hired.

Appendix B (Sydney)

Participant 2 Interview Responses

Gender	Age Range	Connection to STEM	Applied Using Automated System
Woman	25-34	Previously worked in a STEM field	Yes

Question 1: How do you think the use of AI or automated systems in hiring affects fairness for women applying to STEM jobs?

Participant 2: Yes. It's possible that they screen for the "female" selection under the voluntary disclosures, given that increased women in pools of applicants is considered a type of DEI.

Question 2: Have you ever experienced or observed bias during an online or technology-based hiring process (such as resume screen tools, automated assessments, or video interviews)? If so, how did it affect the outcome?

Participant 2: Yes. The most common bias I've seen is asking for a certain amount of publications or unrealistic experience expectations, additionally some have bias against external applicants, as they have a person already in mind. I'm not sure if it's directly resulted in my being no longer considered for roles.

Question 3: Have you ever felt discouraged from applying to a STEM job because the hiring process seemed automated or based on AI tools? Why or why not?

Participant 2: Occasionally if an application/interview process is AI based I will stop my progression. If the system that replies back to me for interviews is too impersonal (automated) with no real person's name attached I typically stop responding. It just doesn't seem like they care about me as a scientist that way.

Question 4: What concerns, if any, do you have about how AI systems make hiring decisions?

Participant 2: AI definitely screens my resume/application for key words that I feel may or may not actually be related to the position. I've probably been thrown out of a pool because the AI screen didn't see a buzz word, even though I know that I would be able to do the job based on my qualifications and the posting. I'm concerned about the AI systems making the process too impersonal. If a real-life person reviews my application, I would assume they have some STEM training and would be able to better gauge me based on the skills (ie they could read between the lines of my skillset and look for follow up questions, or infer how i could fit into the position).

Question 5: What factors do you think AI systems rely on when evaluating candidates, and how might those factors impact women in STEM?

Participant 2: I think the AI algorithm is trained on mainly key words, years experience/age, or what the company wants as a 100% perfect resume. I feel that this can affect all applicants in general, but women in STEM are still considered less and payed less on average than male counterparts nationwide. It's possible some AI can screen based on name and preferentially select against female names in certain disciplines. Im unsure how frequently it does this.

Question 6: How do AI or automated hiring systems influence how you prepare for or approach applying to STEM jobs?

Participant 2: For positions that I really want, I typically also attach a cover letter, even if its generic. I feel like this way a real person MIGHT have to look at it, I can't be sure.

Overall since many applications use some type of AI, I've just given up tailoring my resume each time (because of the sheer volume of applications I've done) and just have to leave it to chance.

Question 7: Do you think AI hiring systems reflect past inequalities in STEM fields, such as underrepresentation of women?

Participant 2: I think that in training the AI review systems its possible bias was introduced unintentionally. The STEM field has in the past been dominated by men, meaning their resumes would inherently be used more often than females when training the model.

Question 8: How important do you think it is for companies to be transparent about how their AI hiring systems make decisions?

Participant 2: Very important. Most applications ask for you to opt-in to the auto AI review. But refusing can mean your resume is thrown out because it takes too much manpower to manually review. While being transparent about the use of AI is important, I'd rather know exactly HOW the model screens applicants.

Appendix C (Saria)

Participant 3 Interview Responses

Gender	Age Range	Connection to STEM	Applied Using Automated System
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Woman	18-24	Undergraduate student	Yes
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Question 1: How do you think the use of AI or automated systems in hiring affects fairness for women applying to STEM jobs?

Participant 3: I can only hope that by using AI, biases are avoided like when hiring managers discriminate based on names and how "ethnic" or feminine they are.

Question 2: Have you ever experienced or observed bias during an online or technology-based hiring process (such as resume screen tools, automated assessments, or video interviews)? If so, how did it affect the outcome?

Participant 3: I haven't seen direct bias more often than not I'm asked to fill out screenings that ask about my gender, ethnicity, and disability status. Although they say those things aren't considered when hiring, I can only imagine how honest that promise is.

Question 3: Have you ever felt discouraged from applying to a STEM job because the hiring process seemed automated or based on AI tools? Why or why not?

Participant 3: If anything I look forward to it because I know it would give me a faster response, if the company uses AI to hire quicker then I can use AI to get hired quicker.

Question 4: What concerns, if any, do you have about how AI systems make hiring decisions?

Participant 3: I have concerns that they view all my accomplishments at face value instead of understanding what it took to gain all the experience I did.

Question 5: What factors do you think AI systems rely on when evaluating candidates, and how might those factors impact women in STEM?

Participant 3: AI systems may dislike gaps in resumes which could disadvantage women who stop working to have families.

Question 6: How do AI or automated hiring systems influence how you prepare for or approach applying to STEM jobs?

Participant 3: I use AI vs AI. If companies use AI to review applications then I have the right to save my own time by using AI to write cover letters.

Question 7: Do you think AI hiring systems reflect past inequalities in STEM fields, such as underrepresentation of women?

Participant 3: I can only hope they do, I don't know what companies are telling their AI systems to look out for.

Question 8: How important do you think it is for companies to be transparent about how their AI hiring systems make decisions?

Participant 3: I think it's very important for companies to be transparent about how they use AI for hiring purposes. At the same time, I see how too much transparency can lead to people cheating the system to get interviews.

Appendix D (Chanroop)

Participant 4 Interview Responses

Gender	Age Range	Connection to STEM	Applied Using Automated System
Woman	25-34	Currently working in a STEM field	No

Question 1: How do you think the use of AI or automated systems in hiring affects fairness for women applying to STEM jobs?

Participant 4: AI has the potential to improve fairness by standardizing evaluations and reducing individual human bias. However, if not carefully designed, it can also scale bias more efficiently. For women in STEM, this means fairness depends heavily on how the system is trained, monitored, and updated. It's not inherently fair or unfair—it depends on implementation.

Question 2: Have you ever experienced or observed bias during an online or technology-based hiring process (such as resume screen tools, automated assessments, or video interviews)? If so, how did it affect the outcome?

Participant 4: I haven't personally experienced clear bias in a way I could definitively attribute to technology, but I've observed that automated screenings can sometimes overlook strong candidates whose experience doesn't fit a traditional mold. This can affect who even gets the opportunity to interview, which is a critical stage in the process.

Question 3: Have you ever felt discouraged from applying to a STEM job because the hiring process seemed automated or based on AI tools? Why or why not?

Participant 4: Not necessarily discouraged, but it can feel impersonal. When the process is heavily automated, it sometimes feels like your application is reduced to data points rather than your full story. That said, I still apply because I know there are human decision-makers involved later in the process, and I focus on getting past that initial screening.

Question 4: What concerns, if any, do you have about how AI systems make hiring decisions?

Participant 4: My main concerns are lack of transparency, potential bias from training data, and over-reliance on rigid criteria. AI can miss context—like leadership,

adaptability, or unconventional career paths—that are harder to quantify but still valuable. I also worry about candidates being filtered out too early without a fair chance to demonstrate their abilities.

Question 5: What factors do you think AI systems rely on when evaluating candidates, and how might those factors impact women in STEM?

Participant 4: AI systems often rely on keywords in resumes, educational background, past job titles, and patterns from previously successful hires. They may also analyze gaps in employment or specific technical skills. The issue is that these systems are trained on historical data, which often reflects existing gender imbalances in STEM. As a result, they can unintentionally favor candidates who match traditional (often male-dominated) career paths, potentially disadvantaging women who may have nonlinear paths, career breaks, or different ways of demonstrating impact.

Question 6: How do AI or automated hiring systems influence how you prepare for or approach applying to STEM jobs?

Participant 4: Even though I didn't use AI when applying to my current role, I was very intentional about tailoring my resume to highlight measurable impact and clearly list technical skills. I focused on making my experience easy to "parse," knowing automated systems might scan it first. Today, I'm more aware of how structured and keyword-optimized applications need to be, and I would likely adapt more strategically if applying again.

Question 7: Do you think AI hiring systems reflect past inequalities in STEM fields, such as underrepresentation of women?

Participant 4: Yes, to some extent. AI systems are only as unbiased as the data they're trained on. If historical hiring data reflects underrepresentation of women, those patterns can be reinforced unless actively corrected. Without careful auditing and intentional design, these systems can unintentionally replicate the same disparities they're meant to reduce.

Question 8: How important do you think it is for companies to be transparent about how their AI hiring systems make decisions?

Participant 4: Transparency is extremely important. Candidates deserve to understand how they're being evaluated, especially when decisions are partially automated. It builds trust and accountability, and it also allows companies to be held responsible for identifying and correcting bias. Without transparency, it's difficult to know whether the process is fair.